Confidence Calibration and Temperature Scaling

Multi-benchmark anomaly (OOD) detection analysis for autonomous driving

MSP baseline with temperature scaling | AuPRC higher is better, FPR95 lower is better

1. Overview

This report evaluates the Maximum Softmax Probability (MSP) baseline combined with temperature scaling across three EoMT checkpoints. Each logit map is divided by a temperature T before the softmax (p = softmax(f / T)); the optimal T is fitted by minimising the validation Negative Log-Likelihood (NLL), then MSP is re-evaluated at fixed temperatures and at the fitted best T. Results are reported on five benchmarks: SMIYC RA-21, SMIYC RO-21, FS L&F, FS Static and Road Anomaly.

2. Cityscapes checkpoint

Fitted optimal temperature: **best t = 0.192934**.

Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
		AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
MSP	-	-	-	-	-	-	-	-	-	-	-
MSP (t = 0.5)	-	65.93	34.26	89.72	0.34	17.59	20.12	62.55	28.50	66.84	19.79
MSP (t = 0.75)	-	65.99	34.26	89.73	0.34	17.60	20.12	62.57	28.50	66.93	19.74
MSP (t = 1.1)	-	66.06	34.26	89.74	0.34	17.61	20.12	62.58	28.50	67.01	19.72
MSP (best t)	-	65.72	34.26	89.65	0.34	17.55	20.13	62.43	28.50	66.64	20.01

Note: the NLL-optimal temperature (0.192934) is the weakest variant on the downstream anomaly metrics, while a fixed t = 1.1 is consistently best. This points to a mismatch between the calibration objective and OOD discriminative ability.

3. COCO checkpoint

Fitted optimal temperature: **best t = 1.501235**.

Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
		AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
MSP	-	-	-	-	-	-	-	-	-	-	-
MSP (t = 0.5)	-	38.46	78.24	2.74	99.98	3.78	97.22	4.66	99.50	18.79	92.02

Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
		AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
MSP (t = 0.75)	-	38.65	77.66	2.75	99.98	3.78	97.21	4.66	99.48	18.83	91.90
MSP (t = 1.1)	-	38.73	77.50	2.75	99.98	3.79	97.21	4.67	99.48	18.86	91.84
MSP (best t)	-	38.77	77.44	2.75	99.98	3.79	97.21	4.67	99.47	18.87	91.82

Note: absolute performance stays very poor (FPR95 near 100% on most benchmarks). The severe domain gap of a COCO-only model is a structural limitation that post-hoc logit scaling cannot bridge.

4. Fine-tuned checkpoint

Fitted optimal temperature: **best t = 1.097883** (effectively 1.1).

Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
		AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
MSP	-	-	-	-	-	-	-	-	-	-	-
MSP (t = 0.5)	-	43.43	100.00	59.46	4.56	2.07	33.50	20.51	21.82	49.57	29.39
MSP (t = 0.75)	-	50.12	100.00	64.59	3.89	3.53	32.78	34.27	21.00	59.55	28.06
MSP (t = 1.1)	-	58.39	88.27	72.90	3.18	7.42	32.19	48.97	19.83	68.52	26.02
MSP (best t)	-	58.36	88.30	72.85	3.18	7.40	32.19	48.91	19.84	68.48	26.03

Note: raising t from 0.5 to 1.1 yields a large, stepwise improvement (e.g. FS Static AuPRC 20.51% to 48.97%). Since the fitted best t is essentially 1.1, a mild over-smoothing of over-confident softmax outputs restores sensitivity to uncertain regions.

5. Key takeaways

- **Calibration overfitting** — minimising validation NLL does not translate into better rare-object or OOD detection.
- **Temperature scaling boundaries** — logit manipulation cannot bridge a severe domain mismatch when the underlying representations are not aligned to the target domain.
- **Smoothing helps fine-tuned models** — a slightly higher temperature (around 1.1) acts as effective regularisation and maximises OOD precision.